

PRAGYA AGARWAL

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PROFESSIONAL SUMMARY

Results-oriented Full-Stack Engineer with 5+ years of experience delivering scalable, secure, and high-performance applications. Expert in building end-to-end solutions—UI/UX, API integration, backend logic, and database design. Known for clean architecture, agile execution, and translating complex requirements into production-ready systems.

TECHNICAL SKILLS

- **Languages:** JavaScript, Python, Java, SQL
- **Frontend:** React, D3.js, HTML5, CSS3
- **Backend:** Django REST Framework, Spring
- **Testing:** Tavern, Junit
- **Tools & DevOps:** GitHub, Jenkins, Docker, Kubernetes, Postman, Swagger
- **Cloud & Databases:** AWS, Azure, MySQL, SQL Server, Cassandra

WORK EXPERIENCE

Senior Software Engineer | Hewlett Packard Enterprise, Texas, US | November 2023 – Present

- **Built the entire frontend** of Manticore Console from the ground up using **React, D3.js**, and **Python/DRF**, delivering 3 project-critical dashboards used by users, admins, and owners.
- **Developed 15+ modular, production-grade components**, using **React Hooks, Context API**, dynamic rendering, and robust **API integrations** with real-time data.
- **Mastered performance** by optimizing re-renders, enabling **lazy loading**, and cutting initial page load times by **30%**.
- **Applied full accessibility (WCAG)**, responsive layouts, **cross-browser testing**, and adhered to the **Frontend Design System**, ensuring a pixel-perfect experience across devices and platforms.
- **Engineered seamless data export flows** using React and backend APIs—eliminated manual reporting for ops and support teams, saving hours per week.
- **Built a complete testing suite** using **Tavern (Python)** and integrated it into CI/CD, covering 100+ API edge cases and **raising release confidence** across the stack.
- **Collaborated** cross-functionally to deliver a cohesive, accessible, and maintainable frontend architecture.
- **Mentored team members** in **Figma, component architecture**, and **testing**, creating scalable patterns and setting quality benchmarks for frontend code.
- **Delivered 10+ live demos**, earned multiple recognition and consistently led with initiative, quality, and user-first thinking.

Research Intern | The University of Texas at Arlington, Texas, US | July 2023 – Present

- **Investigated** security vulnerabilities in over **50 Java Libraries and Frameworks**, to ensure software integrity.
- **Devised** mitigation strategies for **20+ vulnerabilities** to strengthen system robustness and long-term application resilience.

Software Engineer Intern | Copart Inc., Texas, US | May 2022 – August 2022

- **Developed** responsive single-page applications for the finance team using Java, Spring Boot, Angular, and TypeScript, enhancing UI responsiveness and user experience by 70%.
- **Migrated** car seller datasets from Cassandra to MariaDB using ETL and Python scripts. Slashed page response time from 3.125 seconds to 2.5 seconds, ensuring high performance.
- **Enhanced** the car auction portal's system throughput by 85% and **optimized** search efficiency using Kafka and Solr.
- **Improved** code quality by 45% using JUnit for rigorous testing. Deployed code on Azure using Maven, and GitHub.

Software Development Engineer | DXC Technology, Uttar Pradesh, India | January 2018 – July 2021

- **Programmed** scalable back-end Java microservices with Spring Boot for Aviva Life Insurance, attaining a 99.8% **fault tolerance** and effectively managing load distribution even during high traffic surges.
- **Integrated** RESTful APIs using Spring Cloud and incorporated OAuth2 tokens and JWT for secure authentication. Leveraged Java multithreading to process asynchronous requests, boosting system responsiveness by 30%.
- **Refactored** the front end of the insurance application according to client requirements using HTML, CSS, Bootstrap, JavaScript, JSON, and ReactJS. This resulted in a 40% increase in user retention and 60% fewer manual errors.
- **Spearheaded** an Agile team of 5, delivering 50+ applications on AWS using Docker, Kubernetes, Jenkins, and GitHub. This streamlined CI/CD process led to 75% faster deployments and a 35% increase in productivity.
- **Achieved** 100% client satisfaction by resolving urgent production issues utilizing SQL DDL, DML scripts, and Stored Procedures.
- **Mentored** 20 associates in business and technical skills, accelerating onboarding by 65% and enabling quick work assimilation.

EDUCATION

Master of Science, Computer Science (GPA: 3.8/4.0) | The University of Texas at Arlington | August 2021- May 2023

Bachelor of Technology, Computer Science | Dr. A. P. J. Abdul Kalam Technical University | August 2013 - May 2017